MASSACHUSETTS INSTITUTE of TECHNOLOGY
Department of Physics
8.01

Guide to Understanding Line Integrals

1 Introduction

The following math and physics review module on work and line integrals is designed to help you understand the mathematical meaning of line integrals. In particluar, why it is the correct mathematical concept to describe the work done by a non-uniform force (examples: electrical or gravitational forces) acting on an object as it moves along an arbitrary path .

Outline of the module:

1. Review of the scalar dot product and the work done by a constant force.

2. Work done by a non-uniform force for a one dimensional motion. There are two relevant examples, work done. by a spring force and work done by an electrical force, that illustrate how to set up the work definite integral.

3. Work done b y a non-uniform force along an arbitrary path. This section includes a definition of the work line integral.

4. Two examples of the work line integrals are worked out in detail, work done by a uniform gravitational force and a non-uniform gravitational force.

5. The final example is the electric potential difference between two points due to a point-like charged source. This example is particularly important in electrostatics.

2 Work and the Dot Product

The most important physical example of the dot product is work. Recall that when a constant force acts on a mass that is moving along the x-axis, only the component of the force along that direction contributes to the work,

$$W = F_x \Delta x. \tag{1}$$

For example suppose we are pulling a mass along a horizontal surface with a force $\vec{\mathbf{F}}$. Choose coordinates such that horizontal direction is the x-axis and the force $\vec{\mathbf{F}}$ forms an angle β with the positive x-direction. In Figure 1 we show the force vector $\vec{\mathbf{F}} = F_x\,\hat{\mathbf{i}} + F_y\,\hat{\mathbf{j}}$ and the displacement vector $\Delta\vec{\mathbf{x}} = \Delta x\,\hat{\mathbf{i}}$. Note that $\Delta x$is the component of the displacement and hence can be greater, equal, or less than zero.

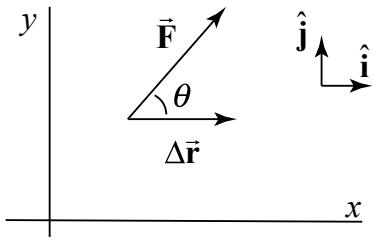


Figure 1: Force and displacement vectors.

Then the dot product between the force vector $\vec{\mathbf{F}}$ and the displacement vector $\Delta\vec{\mathbf{x}}$ is

$$\vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{x}} = (F_x\,\hat{\mathbf{i}} + F_y\,\hat{\mathbf{j}}) \cdot (\Delta x\,\hat{\mathbf{i}}) = F_x\,\Delta x. \tag{2}$$

This is the work done by the force,

$$\Delta W = \vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{x}}. \tag{3}$$

The force $\vec{\mathbf{F}}$ forms an angle β with the positive x-direction. The angle β takes values within the range $-\pi \leq \beta \leq \pi$. Since the x-component of the force is $F_x = F\cos\beta$ where $F = |\mathrm{F}|$denotes the magnitude of $\vec{\mathbf{F}}$, the work done by the force is

$$W = \vec{\mathbf{F}} \cdot \Delta\vec{x} = (F\ \cos\beta)\Delta x. \tag{4}$$

2.1 Example: Work done by a Constant Electric Force

Consider a positive charge$q > 0$ moving between two parallel plates of equal and opposite uniform surface charge density σ, as shown in Figure 2.

Ignoring edge effects, we can use Gauss's Law to calculate the electric field between the plates . The electric field is uniform and points from the positive plate towards the negative plate, $\vec{\mathbf{E}} = E_x\hat{\mathbf{i}}$ with $E_x = \sigma/\varepsilon_0$. Suppose that the charge q has a displacement, $\Delta\vec{\mathbf{x}} = \Delta x\hat{\mathbf{i}}$ with $\Delta x > 0$. The electric force on the charge is

$$\vec{\mathbf{F}}_e = q\vec{\mathbf{E}} = qE_x\hat{\mathbf{i}} = \frac{q\sigma}{\varepsilon_0}\hat{\mathbf{i}}. \tag{5}$$

Because the dot product of a unit vector with itself is one, $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$, the work done by the electric force in displacing the charge $\Delta\vec{\mathbf{x}} = \Delta x\hat{\mathbf{i}}$ is

$$\Delta W = \vec{\mathbf{F}}_e \cdot \Delta\vec{\mathbf{x}} = \frac{q\sigma}{\varepsilon_0}\hat{\mathbf{i}} \cdot \Delta x\hat{\mathbf{i}} = \frac{q\sigma}{\varepsilon_0}\Delta x. \tag{6}$$

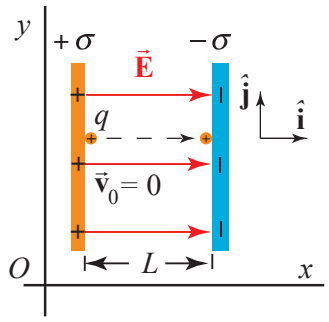


Figure 2: Charge moving in a constant electric field.

The work done on the charge is positive and will increase the kinetic energy of the charge,

$$\Delta W = \Delta K \equiv K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2. \tag{7}$$

If the charge was initially at rest with $v_i = 0$, then the work-kinetic energy law becomes,

$$\frac{q\sigma}{\varepsilon_0}\Delta x = \frac{1}{2}mv_f^2. \tag{8}$$

The speed (magnitude of the velocity) of the charge after it has been displaced $\Delta\vec{\mathbf{x}} = \Delta x\hat{\mathbf{i}}$ is

$$v_f = \sqrt{\frac{2q\sigma}{m\varepsilon_0}\Delta x}. \tag{9}$$

3 Work Non-Constant Force

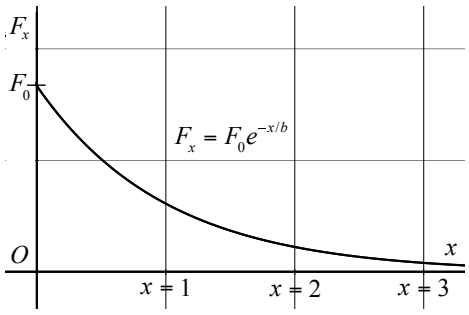


Figure 3: Graph of F_x with respect to the parameter x.

Consider a mass moving in the x-direction under the influence of a non-uniform force that is pointing in the x-direction, $\vec{\mathbf{F}} = F_x\hat{\mathbf{i}}$, where $F_x = F_0 e^{-x/b}$. The mass moves from an initial position x_0 to a final position x_f. In order to calculate the work done by the force, divide up the displacement into a large number N of small displacements Δx_i where the index i marks the ith displacement and takes integer values from 1 to N, with $x_N = x_f$. Let $(F_x)_i$ denote the average value of the x-component of the force in the interval $[x_{i-1}, x_i]$. For the ith displacement we calculate the contribution to the work

$$\Delta W_i = (F_x)_i \Delta x_i. \tag{10}$$

This contribution is a scalar so we add up these scalar quantities to get the total work

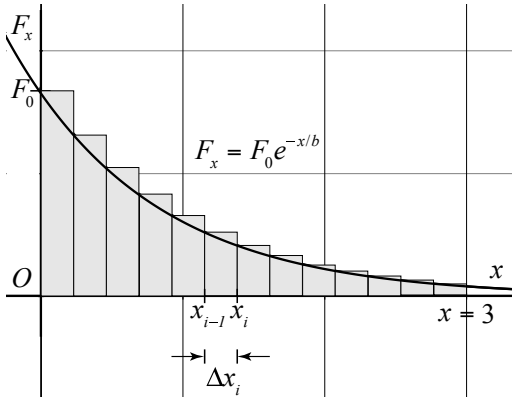


Figure 4: Approximating the area under the graph of F_x with rectangles of width Δx.

$$W_N = \sum_{i=1}^{i=N} \Delta W_i = \sum_{i=1}^{i=N} (F_x)_i \Delta x_i. \tag{11}$$

In order to define a quantity that is independent of the number N of divisions, take the limit as $N \to \infty$ and $|\Delta x_i| \to 0$. Then the work is

$$W = \lim_{\substack{N\to\infty \\ \Delta x_i \to 0}} \sum_{i=1}^{i=N} (F_x)_i \Delta x_i = \int_{x=x_0}^{x=x_f} F_x dx. \tag{12}$$

This last expression is the definition of the integral of the F_x with respect to the parameter x. The work integral is the area under the curve in Figure 3 from $x = x_0$ to $x = x_f$. Let $x_0 = 0$ to a final position $x_f = 3$. With these boundary conditions, the integral in Equation 12 is then

$$W = \int_{x=0}^{x=3} F_0 e^{-x/b} dx = -\frac{F_0}{b}\, e^{-x/b}\Big|_0^3 = \frac{F_0}{b}(1 - e^{-3/b}). \tag{13}$$

3.1 Work done by the Spring Force

Connect one end of an unstretched spring of length l_0 with spring constant k to an object resting on a smooth frictionless table and fix the other end of the spring to a wall. Choose an origin as shown in Figure 5. Stretch the spring by an amount x_i and release the object. How much work does the spring do on the object when the spring is stretched by an amount x_f ?

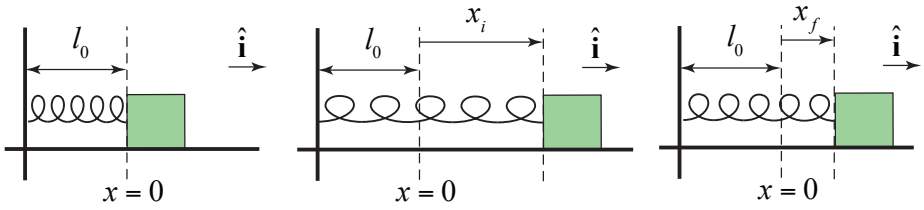


Figure 5: Equilibrium, initial and final states for a spring.

Solution:

We first begin by choosing a coordinate system with our origin located at the position of the object when the spring is unstretched (or uncompressed). We choose the $\hat{\mathbf{i}}$ unit vector to point in the direction the object moves when the spring is being stretched. We choose the coordinate function x to denote the position of the object with respect to the origin. We show the coordinate function and free-body force diagram in Figure 6. The spring force on the object is given by

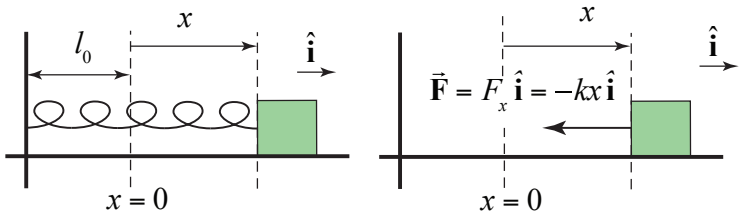


Figure 6: Spring force.

$$\vec{\mathbf{F}} = F_x\,\hat{\mathbf{i}} = -k\,x\,\hat{\mathbf{i}}. \tag{14}$$

In Figure 14 we show the graph of the x-component of the spring force, $F_x\,(x)$, as a function of x.

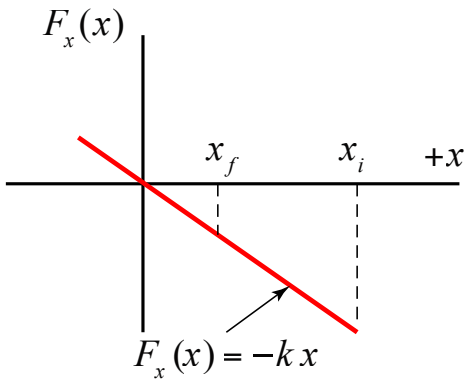


Figure 7: Plot of spring force $F_x\,(x)$ vs. displacement x.

The work done is just the area under the curve for the interval x_i to x_f,

$$W = \int_{x'=x_i}^{x'=x_f} F_x\,(x')dx' = \int_{x'=x_i}^{x'=x_f} -kx'dx' = -\frac{1}{2}k(x_f^2 - x_i^2). \tag{15}$$

This result is independent of the sign of x_i and x_f because both quantities appear as squares. If the spring is less stretched or compressed in the final state than in the initial state, then the absolute value, $|x_f| < |x_i|$, and the work done by the spring force is positive. The spring force does positive work on the body when the spring goes from a state of "greater tension" to a state of "lesser tension."

3.2 Example: Work Done by the Inverse Square Electrical Force

Consider two charges q_1 and q_2 interacting via the electric force. Charge q_1 is fixed in place while charge q_2 is free to move. Choose a coordinate system with the origin at charge q_1. Let r denote the distance between charge q_2 and the origin. Choose the $+\hat{\mathbf{r}}$ unit vector to point away from charge q_1. Initially at time $t = t_0$, the charges are separated by a distance r_0. At time $t = t_f$, the charges are separated by a distance r_f. If the charges have the same sign they will repel and $r_f > r_0$. If the charges have opposite signs, the charges will attract and $r_f < r_0$. How much work does the electric force do on the charge q_2 during either motion?

Solution The electric force between the charges is given by

$$\vec{\mathbf{F}}_e = F_e\hat{\mathbf{r}} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}}. \tag{16}$$

The work done by this electric force on the charge q_2 is the area under the curve in

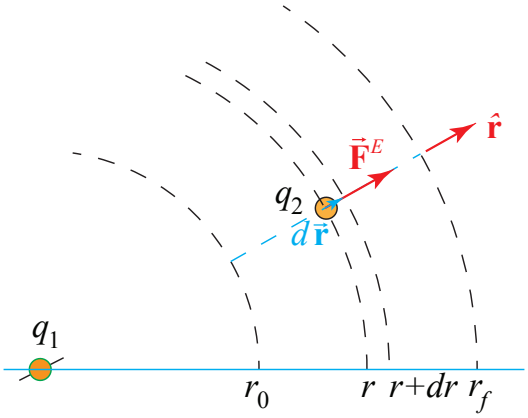


Figure 8: Electric force and infinitesimal displacement of charge q_2.

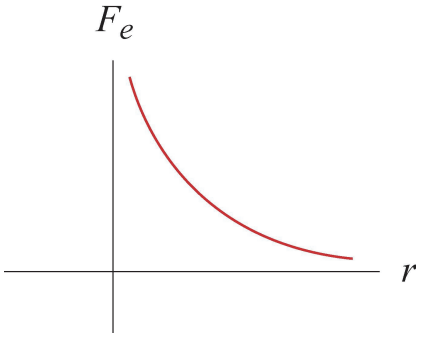


Figure 9: Graph of electric force between the two charges of the same sign as a function of r.

Figure 9 and is given by the integral

$$W = \int_{r_0}^{r_f} F_r dr = \frac{1}{4\pi\varepsilon_0} \int_{r_0}^{r_f} \frac{q_1 q_2}{r^2} dr. \tag{17}$$

Upon evaluation of this integral, we have for the work

$$W = \int_{r_0}^{r_f} \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2} dr = -\frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2}\Big|_{r_0}^{r_f} = -\frac{1}{4\pi\varepsilon_0} q_1 q_2 \left(\frac{1}{r_f} - \frac{1}{r_0}\right). \tag{18}$$

If the charges have the same signs, the charge q_2 will move further from charge q_1, so $r_f > r_0$, and $\frac{1}{r_f} < \frac{1}{r_0}$. Because the charges have the same signs, $q_1 q_2 > 0$. Therefore the work done by electrical force in moving charge q_2 is positive,

$$W = -\frac{1}{4\pi\varepsilon_0} q_1 q_2 (\frac{1}{r_f} - \frac{1}{r_0}) > 0. \tag{19}$$

The charges under the influence of electric forces will naturally move in the direction in which the force does positive work. If the charges have opposite signs, then $q_1 q_2 < 0$. They will attrractl with $r_f < r_0$ and $\frac{1}{r_f} > \frac{1}{r_0}$. Thus the work is once again positive.

4 Work done by a Non-Uniform Force Along an Arbitrary Path

Suppose that a non-uniform force $\vec{\mathbf{F}}$ acts on a point-like body of mass m while the body is moving on a three dimensional curved path. The position vector of the body at time t with respect to a choice of origin is $\vec{\mathbf{r}}(t)$. In Figure 10 we show the path of the body for a time interval $[t_i, t_f]$ moving from an initial position $\vec{\mathbf{r}}_i \equiv \vec{\mathbf{r}}(t = t_i)$ at time $t = t_i$ to a final position $\vec{\mathbf{r}}_f \equiv \vec{\mathbf{r}}(t = t_f)$ at time $t = t_f$.

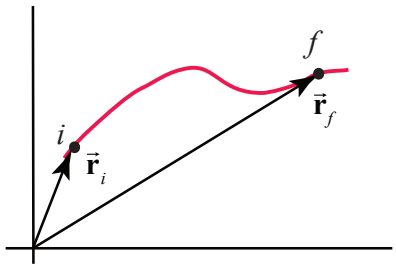


Figure 10: Path traced by the motion of a body.

We divide the time interval $[t_i, t_f]$ into N smaller intervals with $[t_{j-1}, t_j]$, $j = 1, \cdots, N$ with $t_N = t_f$. Consider two position vectors $\vec{\mathbf{r}}_j$ and $\vec{\mathbf{r}}_{j-1}$ the displacement vector during the corresponding time interval as $\Delta\vec{\mathbf{r}}_j = \vec{\mathbf{r}}_j - \vec{\mathbf{r}}_{j-1}$. Let $\vec{\mathbf{F}}$ denote the force acting on the body during the interval $[t_{j-1}, t_j]$. The average force in this interval is $(\vec{\mathbf{F}}_j)_{ave}$ and the average work ΔW_j done by the force during the time interval $[t_{j-1}, t_j]$ is the scalar product between the average force vector and the displacement vector,

$$\Delta W_j = (\vec{\mathbf{F}}_j)_{ave} \cdot \Delta\vec{\mathbf{r}}_j. \tag{20}$$

The force and the displacement vectors for the time interval $[t_{j-1}, t_j]$ are shown in Figure 11.

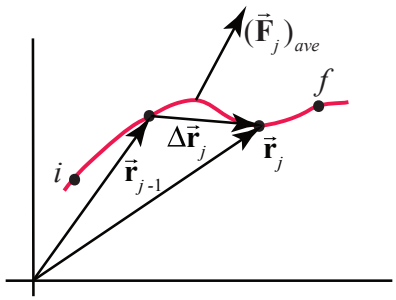


Figure 11: An infinitesimal work element.

The work done is the sum of these scalar contributions for each interval $[t_{j-1}, t_j]$, for $j = 1$ to N,

$$W_N = \sum_{j=1}^{j=N} \Delta W_j = \sum_{j=1}^{j=N} (\vec{\mathbf{F}}_j)_{ave} \cdot \Delta\vec{\mathbf{r}}_j. \tag{21}$$

We would like to define work in a manner that is independent of the way we divide the interval, so we take the limit as $N \to \infty$ and $\left|\Delta\vec{\mathbf{r}}_j\right| \to 0$ for all j. In this limit, as the intervals become smaller and smaller, the distinction between the average force and the actual force vanishes. Thus if this limit exists and is well defined, then the work done by the force is

$$W = \lim_{\substack{N\to\infty \\ \Delta x_j \to 0}} \sum_{j=1}^{j=N} (\vec{\mathbf{F}}_j)_{ave} \cdot \Delta\vec{\mathbf{r}}_j = \int_i^f \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \tag{22}$$

Notice that this summation involves adding scalar quantities. This limit is called the *line integral of the force* $\vec{\mathbf{F}}$. The symbol $d\vec{\mathbf{r}}$ is called the *infinitesimal vector line element*. At time t, $d\vec{\mathbf{r}}$ is tangent to the orbit of the body and is the limit of the displacement vector $\Delta\vec{\mathbf{r}} = \vec{\mathbf{r}}(t + \Delta t) - \vec{\mathbf{r}}(t)$ as Δt approaches zero. In this limit, the parameter t does not appear in the expression in Equation 22.

In general this line integral depends on the particular path the body takes between the initial position $\vec{\mathbf{r}}_i$ and the final position $\vec{\mathbf{r}}_f$, which matters when the force $\vec{\mathbf{F}}$ is non-uniform in space, and when the contribution to the work can vary over different paths in space. We can represent the integral in Equation 22 explicitly in a coordinate system by specifying the infinitesimal vector line element $d\vec{\mathbf{r}}$ and then explicitly computing the scalar product.

4.1 Work Integral in Cartesian Coordinates

In Cartesian coordinates the line element is

$$d\vec{\mathbf{r}} = dx\ \hat{\mathbf{i}} + dy\ \hat{\mathbf{j}} + dz\ \hat{\mathbf{k}}. \tag{23}$$

where dx, dy, and dz represent arbitrary displacements in the $\hat{\mathbf{i}}$-, $\hat{\mathbf{j}}$-, and $\hat{\mathbf{k}}$-directions respectively as seen in Figure 12 The force vector can be represented in vector notation

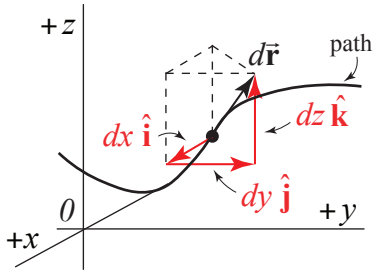


Figure 12: A line element in Cartesian coordinates.

by

$$\vec{\mathbf{F}} = F_x\,\hat{\mathbf{i}} + F_y\,\hat{\mathbf{j}} + F_z\,\hat{\mathbf{k}}. \tag{24}$$

The infinitesimal work is the sum of the work done by the component of the force times the component of the displacement in each direction,

$$dW = F_x dx + F_y dy + F_z dz. \tag{25}$$

Equation 25 is just the scalar product

$$\begin{aligned} dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= (F_x\,\hat{\mathbf{i}} + F_y\,\hat{\mathbf{j}} + F_z\,\hat{\mathbf{k}}) \cdot (dx\,\hat{\mathbf{i}} + dy\,\hat{\mathbf{j}} + dz\,\hat{\mathbf{k}}) \\ &= F_x dx + F_y dy + F_z dz \end{aligned}. \tag{26}$$

The work is

$$W = \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_i}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_i}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} (F_x dx + F_y dy + F_z dz) = \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_i}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} F_x dx + \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_i}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} F_y dy + \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_i}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} F_z dz. \tag{27}$$

4.2 Work Integral in Cylindrical Coordinates

In cylindrical coordinates the line element is

$$d\vec{\mathbf{r}} = dr\,\hat{\mathbf{r}} + r d\theta\,\hat{\theta} + dz\,\hat{\mathbf{k}}. \tag{28}$$

where dr, $d\theta$, and dz represent arbitrary displacements in the $\hat{\mathbf{r}}$-, $\hat{\theta}$-, and $\hat{\mathbf{k}}$-directions respectively as seen in Figure 13 The force vector can be represented in vector notation by

$$\vec{\mathbf{F}} = F_r\,\hat{\mathbf{r}} + F_\theta\,\hat{\theta} + F_z\,\hat{\mathbf{k}}. \tag{29}$$

The infinitesimal work is the scalar product

$$\begin{aligned} dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= (F_r\,\hat{\mathbf{r}} + F_\theta\,\hat{\boldsymbol{\theta}} + F_z\,\hat{\mathbf{k}}) \cdot (dr\,\hat{\mathbf{r}} + r d\theta\,\hat{\boldsymbol{\theta}} + dz\,\hat{\mathbf{k}}) \\ &= F_r dr + F_\theta r d\theta + F_z dz. \end{aligned}. \tag{30}$$

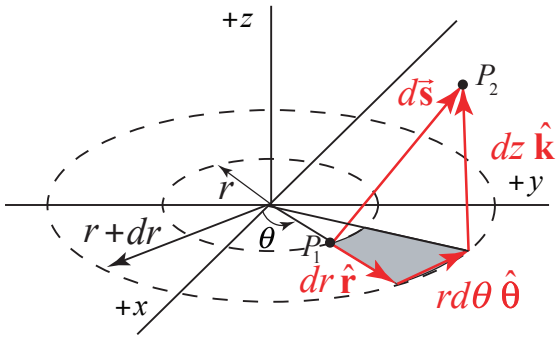


Figure 13: Displacement vector $d\vec{\mathbf{s}}$ between two points.

The work is

$$W = \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_0}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_0}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} (F_r dr + F_\theta r d\theta + F_z dz) = \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_0}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} F_r dr + \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_0}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} F_\theta r d\theta + \int_{\vec{\mathbf{r}}=\vec{\mathbf{r}}_0}^{\vec{\mathbf{r}}=\vec{\mathbf{r}}_f} F_z dz. \tag{31}$$

Another example is the parabolic trajectory associated with projectile motion with the orbit equation in that case is given by

$$y(t) = -\frac{g}{2{v_{x,0}}^2}x^2 + \left(\frac{g x_0}{{v_{x,0}}^2} + \frac{v_{y,0}}{v_{x,0}}\right)x - \frac{v_{y,0}x_0}{v_{x,0}} - \frac{g{x_0}^2}{2{v_{x,0}}^2} + y_0. \tag{32}$$

The difficulty in calculating line integrals is to explicitly determine how the force varies over the trajectory.

4.3 Example: Work Done in a Constant Gravitational Field

The work done in a uniform gravitational field is a fairly straightforward calculation when the mass moves in the direction of the field. However suppose the mass is moving under the influence of gravity, $\vec{\mathbf{F}} = -mg\hat{\mathbf{j}}$ along a parabolic curve given by the equation

$$y = ax - bx^2 + c. \tag{33}$$

The mass begins at the point (x_0, y_0) and ends at the point (x_f, y_f). What is the work done by the gravitational force on the mass?

We begin by introducing a parameter for the path. Let $t = x$. Note that the variable t can take on any value in some interval $[a, b]$. Then

$$y = at - bt^2 + c. \tag{34}$$

The path is now specified by the position vector

$$\vec{\mathbf{r}}(t) = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} = t\hat{\mathbf{i}} + (at - bt^2 + c)\hat{\mathbf{j}}. \tag{35}$$

The infinitesimal line element $d\vec{\mathbf{r}} = (d\vec{\mathbf{r}}/dt)dt$ is tangent to the curve and is therefore

$$d\vec{\mathbf{r}} = dt\,\hat{\mathbf{i}} + (a - 2bt)dt\,\hat{\mathbf{j}}. \tag{36}$$

So the dot product that appears in the line integral can now be calculated,

$$\vec{\mathbf{F}} \cdot d\,\vec{\mathbf{r}} = \vec{\mathbf{F}} \cdot \frac{d\vec{\mathbf{r}}}{dt}dt = -mg\hat{\mathbf{j}} \cdot \left[dt\,\hat{\mathbf{i}} + (a - 2bt)\,dt\,\hat{\mathbf{j}}\right] = -mg(a - 2bt)dt. \tag{37}$$

The parameter t varies from x_0 to x_f. Thus the work integral is

$$W = \int_{\mathrm{r}_0}^{\mathrm{r}_f} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{t=x_0}^{t=x_f} -mg(a - 2bt)dt. \tag{38}$$

This integral is fairly straightforward and after integration the work is

$$\begin{aligned} W &= \int_{x_0}^{x_f} -mg(a - 2bt)dt = \left(- mgat + mgbt^2)\right|_{x_0}^{x_f} \\ &= mg\left[b({x_f}^2 - {x_0}^2) - a(x_f - x_0)\right] \end{aligned}. \tag{39}$$

Since the mass always follows the equation of the parabola,

$$y_0 = ax_0 - b{x_0}^2 + c. \tag{40}$$

$$y_f = ax_f - b{x_f}^2 + c \tag{41}$$

Using these two equations and a little bit of algebra, we have that

$$b({x_f}^2 - {x_0}^2) - a(x_f - x_0) = -(y_f - y_0). \tag{42}$$

The work done by the uniform gravitational force is

$$W = \int_{\vec{\mathrm{r}}_0}^{\vec{\mathrm{r}}_f} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = -mg(y_f - y_0). \tag{43}$$

This result is not surprising since the force is only in the y-direction. Therefore the only non-zero contribution to the work integral is in the y-direction with the result that

$$W = \int_{\mathrm{r}_0}^{\mathrm{r}_f} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{y=y_0}^{y=y_f} F_y dy = \int_{y=y_0}^{y=y_f} -mg\,dy = -mg(y_f - y_0). \tag{44}$$

4.4 Inverse Square Gravitation Force

Consider a system consisting of two objects of masses m_1 and m_2 that are separated by a center-to-center distance $r_{2,1}$. The gravitational force on object 1 due to the interaction between the two objects is given by

$$\vec{\mathbf{F}}^G = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}}. \tag{45}$$

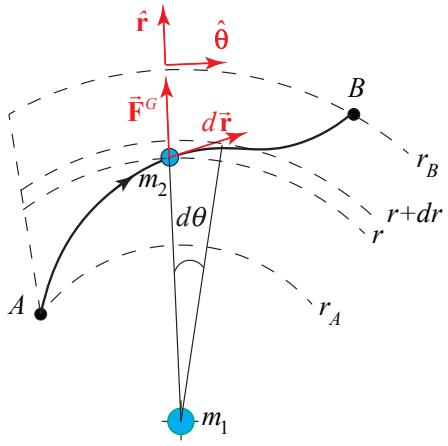


Figure 14: Graph of electric force between the two charges as a function of r.

Assume that the mass of object 1 is much greater than the mass of object 2. Hence treat object 1 as fixed at the center of a polar coordinate system and object 2 is moving along the path from point A to point B under the action of a gravitational force. The infinitesimal displacement vector is

$$d\vec{\mathbf{s}} = dr\hat{\mathbf{r}} + rd\theta\hat{\theta} \tag{46}$$

The scalar product between the gravitational force and the displacement vector is

$$\vec{\mathbf{F}}^G \cdot d\vec{\mathbf{s}} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}} \cdot (dr\,\hat{\mathbf{r}} + rd\theta\hat{\theta}) = -\frac{Gm_1m_2}{r^2}dr. \tag{47}$$

Recall that in polar coordinates $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\theta} \cdot \hat{\theta} = 1; \quad \hat{\mathbf{r}} \cdot \hat{\theta} = 0$. The work done by the gravitational force acting on object 2 is then

$$W = \int_{r_A}^{r_B} \vec{\mathbf{F}}^G \cdot d\vec{\mathbf{s}} = \int_{r_A}^{r_B} \left(-\frac{Gm_1m_2}{r^2}\right) dr = -Gm_1m_2\left(\frac{1}{r_B} - \frac{1}{r_A}\right). \tag{48}$$

4.5 Example: Electric Potential Difference due to a Point Charge

When a point-like charged object with charge q is placed in an electric field $\vec{\mathbf{E}}$, there is an electric force acting on the charge:

$$\vec{\mathbf{F}}_q = q\vec{\mathbf{E}} \tag{49}$$

The work done by the electric force in moving the charged object from A to B is the line integral

$$W_{AB} = \int_A^B q_t\vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} \tag{50}$$

The *electric potential difference* is defined to be the negative of the work done per charge in moving a *test charged object* with charge q_t from A to B

$$\Delta V_{AB} = \frac{-W_{AB}}{q_t} = -\int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} \tag{51}$$

Now consider a point-like charged object with charge $q_s > 0$ that is held fixed at the origin of a polar coordinate system. This object is the *source of the electric field.* Consider an arbitrary point P with coordinates (r, θ) and a set of unit vectors at the point P, $(\hat{\mathbf{r}}, \hat{\theta})$, (Figure 15). The electric field at the point P due to the source charge is given by

$$\vec{\mathbf{E}}_s(P) = \frac{1}{4\pi\varepsilon_0}\frac{q_s}{r^2}\hat{\mathbf{r}}. \tag{52}$$

What is the electric potential difference between the point B, which is located a distance

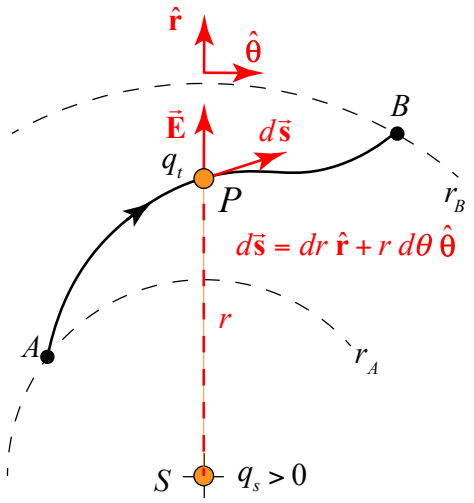


Figure 15: Coordinate system for calculating electric potential difference $V_B - V_A$.

r_B from the source charge and the point A which is located a distance r_A?

Solution:

The electric potential difference is given by the integral

$$\Delta V_{AB} \equiv V_B - V_A = -\int_A^B \vec{\mathbf{E}}_s(P) \cdot d\vec{\mathbf{s}}\,, \tag{53}$$

where $d\vec{\mathbf{s}}$ is the infinitesimal displacement element at the point P located a distance r from the source charge on the path from the initial point A to the final point B. In polar

coordinates $d\vec{\mathbf{s}} = dr\ \hat{\mathbf{r}} + rd\theta\ \hat{\boldsymbol{\theta}}$. We can calculate the dot product in the integrand and find that

$$\vec{\mathbf{E}}_s(P) \cdot d\vec{\mathbf{s}} = \frac{1}{4\pi\varepsilon_0}\frac{q_s}{r^2}\hat{\mathbf{r}} \cdot \ (dr\ \hat{\mathbf{r}} + rd\theta\ \hat{\boldsymbol{\theta}}) = \frac{1}{4\pi\varepsilon_0}\frac{q_s}{r^2}dr. \tag{54}$$

Note any displacement in the tangent direction is perpendicular to the electric field so it does not contribute to the integral. The limits of the integral are r_A and r_B. We can now calculate this integral and find that

$$V(B) - V(A) = -\frac{q_s}{4\pi\varepsilon_0}\int_{r_A}^{r_B}\frac{dr}{r^2} = \frac{q_s}{4\pi\varepsilon_0}\frac{1}{r}\bigg|_{r_A}^{r_B} = \frac{q_s}{4\pi\varepsilon_0}\left(\frac{1}{r_B} - \frac{1}{r_A}\right) \tag{55}$$

Checking signs: In this example, from the figure $r_B > r_A$ and the source charge is positive charge, $q_s > 0$, then the potential difference is negative . The potential decreases as you move away from a positive charge. If $q_s < 0$, then the potential increases as you move away from a negative charge. Equation (1) is general and holds for all cases $r_B > r_A$, $r_B = r_A$, $r_B < r_A$, $q_s > 0$, and $q_s < 0$.